# Retrieve FC port information

ONTAP 9.11.1 REST API reference

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# Retrieve FC port information

## Network FC logins endpoint overview

### Overview

Fibre Channel (FC) logins represent connections formed by FC initiators that have successfully logged in to ONTAP. This represents the FC login on which higher-level protocols such as Fibre Channel Protocol and NVMe over FC (NVMe/FC) rely.

The Fibre Channel logins REST API provides information about active FC logins.

### Examples

#### Retrieving all FC logins

```
# The API:
GET /api/network/fc/logins

# The call:
curl -X GET "https://<mgmt-ip>/api/network/fc/logins" -H "Accept:
application/hal+json"

# The response:
{
"records": [
  {
    "svm": {
      "uuid": "056403da-83a7-4b13-bc78-6a93e8ea3596",
      "name": "svm1",
      "_links": {
        "self": {
          "href": "/api/svm/svms/056403da-83a7-4b13-bc78-6a93e8ea3596"
        }
      }
    },
    "interface": {
      "uuid": "01056403-1383-bc4b-786a-93e8ea35969d",
      "name": "lif1",
      "_links": {
        "self": {
          "href": "/api/network/fc/interfaces/01056403-1383-bc4b-786a-
93e8ea35969d"
        }
      }
    },
```

```
    "initiator": {
      "wwpn": "8b:21:2f:07:00:00:00:00"
    },
    "_links": {
      "self": {
        "href": "/api/network/fc/logins/01056403-1383-bc4b-786a-
93e8ea35969d/8b%3A21%3A2f%3A07%3A00%3A00%3A00%3A00"
      }
    }
  },
  {
    "svm": {
      "uuid": "056403da-83a7-4b13-bc78-6a93e8ea3596",
      "name": "svm1",
      "_links": {
        "self": {
          "href": "/api/svm/svms/056403da-83a7-4b13-bc78-6a93e8ea3596"
        }
      }
    },
    "interface": {
      "uuid": "02056403-1383-bc4b-786a-93e8ea35969d",
      "name": "lif2",
      "_links": {
        "self": {
          "href": "/api/network/fc/interfaces/02056403-1383-bc4b-786a-
93e8ea35969d"
        }
      }
    },
    "initiator": {
      "wwpn": "8c:21:2f:07:00:00:00:00"
    },
    "_links": {
      "self": {
        "href": "/api/network/fc/logins/02056403-1383-bc4b-786a-
93e8ea35969d/8c%3A21%3A2f%3A07%3A00%3A00%3A00%3A00"
      }
    }
  },
  {
    "svm": {
      "uuid": "156403da-83a7-4b13-bc78-6a93e8ea3596",
      "name": "svm2",
      "_links": {
        "self": {
```

```
          "href": "/api/svm/svms/156403da-83a7-4b13-bc78-6a93e8ea3596"
        }
      }
    },
    "interface": {
      "uuid": "03056403-1383-bc4b-786a-93e8ea35969d",
      "name": "lif3",
      "_links": {
        "self": {
          "href": "/api/network/fc/interfaces/00056403-1383-bc4b-786a-
93e8ea35969d"
        }
      }
    },
    "initiator": {
      "wwpn": "8a:21:2f:07:00:00:00:00"
    },
    "_links": {
      "self": {
        "href": "/api/network/fc/logins/00056403-1383-bc4b-786a-
93e8ea35969d/8a%3A21%3A2f%3A07%3A00%3A00%3A00%3A00"
      }
    }
  }
],
"num_records": 3,
"_links": {
  "self": {
    "href": "/api/network/fc/logins"
  }
}
}
```


---

#### Retrieving all FC logins with data protocol *fcp* in SVM *svm1*

The `svm.name` and `protocol` query parameters are used to perform the query.

```
# The API:
GET /api/network/fc/logins

# The call:
curl -X GET "https://<mgmt-
ip>/api/network/fc/logins?svm.name=svm1&protocol=fcp" -H "Accept:
application/hal+json"
```

---

```
# The response:
{
"records": [
  {
    "svm": {
      "uuid": "056403da-83a7-4b13-bc78-6a93e8ea3596",
      "name": "svm1",
      "_links": {
        "self": {
          "href": "/api/svm/svms/056403da-83a7-4b13-bc78-6a93e8ea3596"
        }
      }
    },
    "interface": {
      "uuid": "01056403-1383-bc4b-786a-93e8ea35969d",
      "name": "lif2",
      "_links": {
        "self": {
          "href": "/api/network/fc/interfaces/01056403-1383-bc4b-786a-
93e8ea35969d"
        }
      }
    },
    "initiator": {
      "wwpn": "8b:21:2f:07:00:00:00:00"
    },
    "protocol": "fcp",
    "_links": {
      "self": {
        "href": "/api/network/fc/logins/01056403-1383-bc4b-786a-
93e8ea35969d/8b%3A21%3A2f%3A07%3A00%3A00%3A00%3A00"
      }
    }
  },
  {
    "svm": {
      "uuid": "056403da-83a7-4b13-bc78-6a93e8ea3596",
      "name": "svm1",
      "_links": {
        "self": {
          "href": "/api/svm/svms/056403da-83a7-4b13-bc78-6a93e8ea3596"
        }
      }
    },
    "interface": {
```

```
      "uuid": "02056403-1383-bc4b-786a-93e8ea35969d",
      "name": "lif3",
      "_links": {
        "self": {
          "href": "/api/network/fc/interfaces/02056403-1383-bc4b-786a-
93e8ea35969d"
        }
      }
    },
    "initiator": {
      "wwpn": "8c:21:2f:07:00:00:00:00"
    },
    "protocol": "fcp",
    "_links": {
      "self": {
        "href": "/api/network/fc/logins/02056403-1383-bc4b-786a-
93e8ea35969d/8c%3A21%3A2f%3A07%3A00%3A00%3A00%3A00"
      }
    }
  }
],
"num_records": 2,
"_links": {
  "self": {
    "href": "/api/network/fc/logins?svm.name=svm1&protocol=fcp"
  }
}
}
```

---

#### Retrieving all FC logins for initiators belonging to *igroup1* and returning all of their properties

The `igroups.name` query parameter is used to perform the query. The `fields` query parameter is used to return all of the properties.

```
# The API:
GET /api/network/fc/logins

# The call:
curl -X GET "https://<mgmt-
ip>/api/network/fc/logins?igroups.name=igroup1&fields=*" -H "Accept:
application/hal+json"

# The response:
{
```

---

```
"records": [
  {
    "svm": {
      "uuid": "056403da-83a7-4b13-bc78-6a93e8ea3596",
      "name": "svm1",
      "_links": {
        "self": {
          "href": "/api/svm/svms/056403da-83a7-4b13-bc78-6a93e8ea3596"
        }
      }
    },
    "interface": {
      "uuid": "01056403-1383-bc4b-786a-93e8ea35969d",
      "name": "lif2",
      "wwpn": "8b:21:2f:07:00:00:00:00",
      "_links": {
        "self": {
          "href": "/api/network/fc/interfaces/01056403-1383-bc4b-786a-
93e8ea35969d"
        }
      }
    },
    "initiator": {
      "wwpn": "8b:21:2f:07:00:00:00:00",
      "wwnn": "95:21:2f:07:00:00:00:00",
      "comment": "Example information about this initiator"
    },
    "igroups": [
      {
        "uuid": "243bbb8a-46e9-4b2d-a508-a62dc93df9d1",
        "name": "igroup1",
        "_links": {
          "self": {
            "href": "/api/protocols/san/igroups/243bbb8a-46e9-4b2d-a508-
a62dc93df9d1"
          }
        }
      }
    ],
    "port_address": "8aa53",
    "protocol": "fcp",
    "_links": {
      "self": {
        "href": "/api/network/fc/logins/01056403-1383-bc4b-786a-
93e8ea35969d/8b%3A21%3A2f%3A07%3A00%3A00%3A00%3A00"
      }
```

```
      }
    }
  ],
  "num_records": 1,
  "_links": {
    "self": {
      "href": "/api/network/fc/logins?igroups.name=igroup1&fields=*"
    }
  }
}
```


# Retrieve FC logins

GET `/network/fc/logins`

**Introduced In:** 9.6

Retrieves FC logins.

## Related ONTAP commands

- `vserver fcp initiator show`

## Learn more

- SAN: DOC /network/fc/logins
- NVMe: DOC /network/fc/logins

## Parameters

| Name | Type | In | Required | Description |
|---|---|---|---|---|
| protocol | string | query | False | Filter by protocol |
| svm.uuid | string | query | False | Filter by svm.uuid |
| svm.name | string | query | False | Filter by svm.name |
| igroups.name | string | query | False | Filter by igroups.name<br>• maxLength: 96<br>• minLength: 1 |
| igroups.uuid | string | query | False | Filter by igroups.uuid |

| Name | Type | In | Required | Description |
| --- | --- | --- | --- | --- |
| initiator.port_address | string | query | False | Filter by initiator.port_address |
| initiator.wwpn | string | query | False | Filter by initiator.wwpn |
| initiator.comment | string | query | False | Filter by initiator.comment<br>• Introduced in: 9.9 |
| initiator.wwnn | string | query | False | Filter by initiator.wwnn |
| initiator.aliases | string | query | False | Filter by initiator.aliases |
| interface.name | string | query | False | Filter by interface.name |
| interface.wwpn | string | query | False | Filter by interface.wwpn |
| interface.uuid | string | query | False | Filter by interface.uuid |
| fields | array[string] | query | False | Specify the fields to return. |
| max_records | integer | query | False | Limit the number of records returned. |
| return_records | boolean | query | False | The default is true for GET calls. When set to false, only the number of records is returned.<br>• Default value: 1 |

| Name | Type | In | Required | Description |
|---|---|---|---|---|
| return_timeout | integer | query | False | The number of seconds to allow the call to execute before returning. When iterating over a collection, the default is 15 seconds. ONTAP returns earlier if either max records or the end of the collection is reached.<br>• Default value: 1<br>• Max value: 120<br>• Min value: 0 |
| order_by | array[string] | query | False | Order results by specified fields and optional [asc |

## Response

```
Status: 200, Ok
```

| Name | Type | Description |
|---|---|---|
| _links | _links | |
| num_records | integer | The number of records in the response. |
| records | array[fc_login] | |

**Example response**


```
{
  "_links": {
    "next": {
      "href": "/api/resourcelink"
    },
    "self": {
      "href": "/api/resourcelink"
    }
  },
  "num_records": 1,
  "records": {
    "_links": {
      "next": {
        "href": "/api/resourcelink"
      },
      "self": {
        "href": "/api/resourcelink"
      }
    },
    "igroups": {
      "_links": {
        "self": {
          "href": "/api/resourcelink"
        }
      },
      "name": "igroup1",
      "uuid": "4ea7a442-86d1-11e0-ae1c-123478563412"
    },
    "initiator": {
      "aliases": {
      },
      "comment": "This is an FC initiator for host 5",
      "port_address": "5060A",
      "wwnn": "2f:a0:00:a0:98:0b:56:13",
      "wwpn": "2f:a0:00:a0:98:0b:56:13"
    },
    "interface": {
      "_links": {
        "self": {
          "href": "/api/resourcelink"
        }
      },
      "name": "fc_lif1",
      "uuid": "3a09ab42-4da1-32cf-9d35-3385a6101a0b",
```

```
      "wwpn": "20:00:00:50:56:b4:13:a8"
    },
    "protocol": "fc_nvme",
    "svm": {
      "_links": {
        "self": {
          "href": "/api/resourcelink"
        }
      },
      "name": "svm1",
      "uuid": "02c9e252-41be-11e9-81d5-00a0986138f7"
    }
  }
}
```


## Error

```
Status: Default, Error
```

| Name | Type | Description |
|---|---|---|
| error | error | |

### Example error


```
{
  "error": {
    "arguments": {
      "code": "string",
      "message": "string"
    },
    "code": "4",
    "message": "entry doesn't exist",
    "target": "uuid"
  }
}
```


## Definitions

**See Definitions**

href

| Name | Type | Description |
|---|---|---|
| href | string | |

_links

| Name | Type | Description |
|---|---|---|
| next | href | |
| self | href | |

_links

| Name | Type | Description |
|---|---|---|
| self | href | |

igroups

| Name | Type | Description |
|---|---|---|
| _links | _links | |
| name | string | The name of the initiator group. |
| uuid | string | The unique identifier of the initiator group. |

initiator

Information about the logged in FC initiator.

| Name | Type | Description |
|---|---|---|
| aliases | array[string] | The logged in initiator world wide port name (WWPN) aliases. |
| comment | string | A comment available for use by the administrator. This is modifiable from the initiator REST endpoint directly. See PATCH /protocols/san/igroups/{igroup.uuid}/initiators/{name} . |

| Name | Type | Description |
| --- | --- | --- |
| port_address | string | The port address of the initiator's FC port.<br><br>Each port in an FC switched fabric has its own unique port address for routing purposes. The port address is assigned by a switch in the fabric when that port logs in to the fabric. This property refers to the address given by a switch to the initiator port.<br><br>This is useful for obtaining statistics and diagnostic information from FC switches.<br><br>This is a hexadecimal encoded numeric value. |
| wwnn | string | The logged in initiator world wide node name (WWNN). |
| wwpn | string | The logged in initiator WWPN. |

interface

An FC interface.

| Name | Type | Description |
| --- | --- | --- |
| _links | _links | |
| name | string | The name of the FC interface. |
| uuid | string | The unique identifier of the FC interface. |
| wwpn | string | The WWPN of the FC interface. |

svm

| Name | Type | Description |
| --- | --- | --- |
| _links | _links | |
| name | string | The name of the SVM. |
| uuid | string | The unique identifier of the SVM. |

fc_login

A Fibre Channel (FC) login represents a connection formed by an FC initiator that has successfully logged in to ONTAP. This represents the FC login on which higher-level protocols such as Fibre Channel Protocol and NVMe over Fibre Channel (NVMe/FC) rely.

| Name | Type | Description |
|---|---|---|
| _links | _links | |
| igroups | array[igroups] | The initiator groups in which the initiator is a member. |
| initiator | initiator | Information about the logged in FC initiator. |
| interface | interface | An FC interface. |
| protocol | string | The data protocol used to perform the login. |
| svm | svm | |

error_arguments

| Name | Type | Description |
|---|---|---|
| code | string | Argument code |
| message | string | Message argument |

error

| Name | Type | Description |
|---|---|---|
| arguments | array[error_arguments] | Message arguments |
| code | string | Error code |
| message | string | Error message |
| target | string | The target parameter that caused the error. |

# Retrieve an FC login

GET `/network/fc/logins/{interface.uuid}/{initiator.wwpn}`

**Introduced In:** 9.6

Retrieves an FC login.

## Related ONTAP commands

- `vserver fcp initiator show`

## Learn more

- SAN: DOC /network/fc/logins
- NVMe: DOC /network/fc/logins

## Parameters

| Name | Type | In | Required | Description |
|---|---|---|---|---|
| interface.uuid | string | path | True | The unique identifier of the FC interface through which the initiator logged in. |
| initiator.wwpn | string | path | True | The world wide port name (WWPN) of the initiator. |
| fields | array[string] | query | False | Specify the fields to return. |

## Response

```
Status: 200, Ok
```

| Name | Type | Description |
|---|---|---|
| _links | _links | |
| igroups | array[igroups] | The initiator groups in which the initiator is a member. |
| initiator | initiator | Information about the logged in FC initiator. |
| interface | interface | An FC interface. |
| protocol | string | The data protocol used to perform the login. |

| Name | Type | Description |
| --- | --- | --- |
| svm | svm | |

**Example response**


```
{
  "_links": {
    "next": {
      "href": "/api/resourcelink"
    },
    "self": {
      "href": "/api/resourcelink"
    }
  },
  "igroups": {
    "_links": {
      "self": {
        "href": "/api/resourcelink"
      }
    },
    "name": "igroup1",
    "uuid": "4ea7a442-86d1-11e0-ae1c-123478563412"
  },
  "initiator": {
    "aliases": {
    },
    "comment": "This is an FC initiator for host 5",
    "port_address": "5060A",
    "wwnn": "2f:a0:00:a0:98:0b:56:13",
    "wwpn": "2f:a0:00:a0:98:0b:56:13"
  },
  "interface": {
    "_links": {
      "self": {
        "href": "/api/resourcelink"
      }
    },
    "name": "fc_lif1",
    "uuid": "3a09ab42-4da1-32cf-9d35-3385a6101a0b",
    "wwpn": "20:00:00:50:56:b4:13:a8"
  },
  "protocol": "fc_nvme",
  "svm": {
    "_links": {
      "self": {
        "href": "/api/resourcelink"
      }
    },
    "name": "svm1",
```

```
    "uuid": "02c9e252-41be-11e9-81d5-00a0986138f7"
  }
}
```

## Error

```
Status: Default
```

ONTAP Error Response Codes

| Error Code | Description |
|---|---|
| 4 | The Fibre Channel login specified does not exist. |
| 5373983 | An invalid WWPN was supplied. |
| 5374881 | The Fibre Channel interface specified does not exist. |

| Name | Type | Description |
|---|---|---|
| error | error | |

**Example error**

```
{
  "error": {
    "arguments": {
      "code": "string",
      "message": "string"
    },
    "code": "4",
    "message": "entry doesn't exist",
    "target": "uuid"
  }
}
```

## Definitions

**See Definitions**

href

| Name | Type | Description |
|---|---|---|
| href | string | |

_links

| Name | Type | Description |
|---|---|---|
| next | href | |
| self | href | |

_links

| Name | Type | Description |
|---|---|---|
| self | href | |

igroups

| Name | Type | Description |
|---|---|---|
| _links | _links | |
| name | string | The name of the initiator group. |
| uuid | string | The unique identifier of the initiator group. |

initiator

Information about the logged in FC initiator.

| Name | Type | Description |
|---|---|---|
| aliases | array[string] | The logged in initiator world wide port name (WWPN) aliases. |
| comment | string | A comment available for use by the administrator. This is modifiable from the initiator REST endpoint directly. See PATCH /protocols/san/igroups/{igroup.uuid}/initiators/{name} . |

| Name | Type | Description |
| --- | --- | --- |
| port_address | string | The port address of the initiator's FC port.<br><br>Each port in an FC switched fabric has its own unique port address for routing purposes. The port address is assigned by a switch in the fabric when that port logs in to the fabric. This property refers to the address given by a switch to the initiator port.<br><br>This is useful for obtaining statistics and diagnostic information from FC switches.<br><br>This is a hexadecimal encoded numeric value. |
| wwnn | string | The logged in initiator world wide node name (WWNN). |
| wwpn | string | The logged in initiator WWPN. |

interface

An FC interface.

| Name | Type | Description |
| --- | --- | --- |
| _links | _links | |
| name | string | The name of the FC interface. |
| uuid | string | The unique identifier of the FC interface. |
| wwpn | string | The WWPN of the FC interface. |

svm

| Name | Type | Description |
| --- | --- | --- |
| _links | _links | |
| name | string | The name of the SVM. |
| uuid | string | The unique identifier of the SVM. |

error_arguments

| Name | Type | Description |
| --- | --- | --- |
| code | string | Argument code |
| message | string | Message argument |

error

| Name | Type | Description |
| --- | --- | --- |
| arguments | array[error_arguments] | Message arguments |
| code | string | Error code |
| message | string | Error message |
| target | string | The target parameter that caused the error. |

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